

Scalar product

Basic skills

Q1. a)

$$\begin{pmatrix} 5 \\ 3 \\ 6 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = 5 \times 1 + 3 \times 0 + 6 \times 2 = 17$$

b) $\begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} -4 \\ 2 \\ -1 \end{pmatrix} = -1 \times -4 + 0 \times 2 + 3 \times -1 = 1$

c) $\begin{pmatrix} 3 \\ 1 \\ -4 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix} = 3 - 2 + 4 = 5$

Q2. a)

For magnitude it's Pythagoras

$$|\underline{a}| = \sqrt{3^2 + 4^2 + 5^2} = \sqrt{50} = 5\sqrt{2}$$

b) $|\underline{b}| = \sqrt{(-1)^2 + 2^2 + (-5)^2} = \sqrt{30}$

c) $|\underline{c}| = \sqrt{(-1)^2 + (-2)^2 + (-5)^2} = \sqrt{30}$

magnitude has no care about direction

Q3. a) For angle need scalar Product and magnitude

$$\underline{a} = \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix} \quad \underline{b} = \begin{pmatrix} 2 \\ 1 \\ 8 \end{pmatrix} \quad \underline{a} \cdot \underline{b} = 6 + 4 + 40 = 50$$

$$|\underline{a}| = \sqrt{50} \quad |\underline{b}| = \sqrt{69}$$

$$\text{so } \cos \theta = \frac{50}{\sqrt{50}\sqrt{69}} = \frac{50}{\sqrt{3450}} = \frac{10}{\sqrt{138}}$$

$$\theta = \cos^{-1}\left(\frac{10}{\sqrt{138}}\right) = 31.7^\circ$$

could also give $180 - 31.7^\circ = 148.3^\circ$

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Obtuse

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Obtuse

b)

$$\underline{a} = \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix} \quad \underline{b} = \begin{pmatrix} 10 \\ -2 \\ 4 \end{pmatrix} \quad \text{as } \underline{b} = 2\underline{a}$$

Parallel so 0° between them

could do same as before to get $\theta = 0^\circ$

Competency skills

Q4. $\underline{a} = \begin{pmatrix} 4 \\ -2 \\ 3 \end{pmatrix}$ $\underline{b} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

Position vector

y-axis ($\underline{j}$)

Note. could use scalar multiples of either

Now $\underline{a} \cdot \underline{b} = -2$ $|\underline{a}| = \sqrt{29}$ $|\underline{b}| = 1$

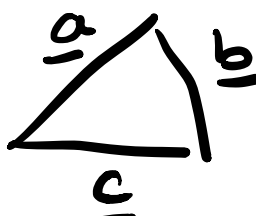
so $\cos \theta = \frac{-2}{\sqrt{29}}$ $\theta = 111.8^\circ$

Q5. $\underline{a} \cdot \underline{b} = 0$ if perpendicular

as $\underline{a} \cdot \underline{b} = 2x + 75 + 8x = 10x + 75$

$10x + 75 = 0$ so $x = -7.5$

Q6. need vectors between points


$$\underline{a} = \begin{pmatrix} 5 \\ -2 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ -4 \\ -2 \end{pmatrix}$$
$$\underline{b} = \begin{pmatrix} 10 \\ 5 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 9 \\ 3 \\ -1 \end{pmatrix}$$

$$\underline{c} = \begin{pmatrix} 10 \\ 5 \\ 2 \end{pmatrix} - \begin{pmatrix} 5 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 7 \\ 2 \end{pmatrix}$$

Now • for a, b

$$\cos \theta = \frac{36 - 12 + 2}{\sqrt{4^2 + 4^2 + 2^2} \sqrt{9^2 + 3^2 + 1^2}} = \frac{\sqrt{91}}{21}$$

$$\theta = 63.0^\circ$$

• for a, c

$$\cos \theta = \frac{20 - 28 - 4}{\sqrt{4^2 + 4^2 + 2^2} \sqrt{5^2 + 7^2 + 2^2}} = \frac{-2}{\sqrt{78}}$$

$$\theta = 103.0^\circ \leftarrow \text{change to } 77^\circ \text{ so add to } 180^\circ$$

• for b, c

$$\cos \theta = \frac{45 + 21 - 2}{\sqrt{9^2 + 3^2 + 1^2} \sqrt{5^2 + 7^2 + 2^2}} = \frac{64}{7\sqrt{143}}$$

$$\theta = 40.0^\circ$$

Q7. $(\underline{i} + \underline{j} + \underline{k}) \cdot (2\underline{i} + \underline{j} + 4\underline{k})$

$$= 1 \times 2 + 1 \times 1 + 1 \times 4 = 7$$

magnitudes $\sqrt{3}$, $\sqrt{21}$

$$\text{so } \cos \theta = \frac{7}{\sqrt{63}} = \frac{7}{3\sqrt{7}}$$

$$\theta = 28.1^\circ$$

$$\text{so } 180^\circ - 28.1^\circ = 151.9^\circ$$

Q8. only $\underline{i} \cdot \underline{i} = 1$, $\underline{j} \cdot \underline{j} = 1$, $\underline{k} \cdot \underline{k} = 1$
all others zero

$$\underbrace{\underline{i} \cdot \underline{i}}_1 + 3 \underbrace{(\underline{j} \cdot \underline{k})}_0 + 8 \underbrace{(\underline{k} \cdot \underline{k})}_1 = 9$$

Q9.

$$\underline{a} \cdot \underline{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

normal $\nearrow = b_1 a_1 + b_2 a_2 + b_3 a_3 = \underline{b} \cdot \underline{a}$

multiplication is commutative

Q10.

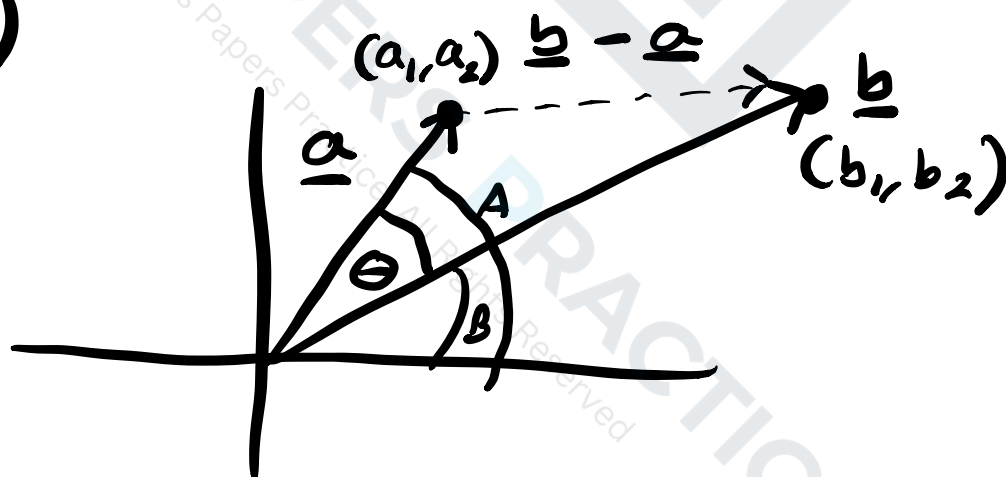
As the angle between $\underline{a}$ and $\underline{a}$ is 0° $\cos \theta = 1$ hence,

$$1 = \frac{\underline{a} \cdot \underline{a}}{|\underline{a}| |\underline{a}|} \text{ so } \underline{a} \cdot \underline{a} = |\underline{a}|^2$$

OR $\underline{a} \cdot \underline{a} = a_1^2 + a_2^2 + a_3^2 = |\underline{a}|^2$

Advanced skills

Q11. a)



b) $\cos(A-B) = \cos A \cos B + \sin A \sin B$

c) $A-B = \theta$

so $\cos A = \frac{a_1}{|\underline{a}|}$ $\cos B = \frac{b_1}{|\underline{b}|}$

$$\sin A = \frac{a_2}{|a|} \quad \sin B = \frac{b_2}{|b|}$$

$$\begin{aligned} \text{Hence } \cos \theta &= \frac{a_1}{|a|} \frac{b_1}{|b|} + \frac{a_2}{|a|} \frac{b_2}{|b|} \\ &= \frac{a_1 b_1 + a_2 b_2}{|a| |b|} = \frac{\underline{a} \cdot \underline{b}}{|a| |b|} \end{aligned}$$

Q12. a)

$$(\underline{a} + \underline{b}) \cdot \underline{c} = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ a_3 + b_3 \end{pmatrix} \cdot \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$$

$$= (a_1 + b_1)c_1 + (a_2 + b_2)c_2$$

$$+ (a_3 + b_3)c_3 = a_1 c_1 + a_2 c_2 + a_3 c_3 + b_1 c_1 + b_2 c_2 + b_3 c_3$$

$$= \underline{a} \cdot \underline{c} + \underline{b} \cdot \underline{c}$$

b)

$$(\underline{a} + \underline{b}) \cdot \underline{b} = \underline{a} \cdot \underline{b} + \underline{b} \cdot \underline{b}$$

$$= |a| |b| \cos \theta + |b|^2$$

$$= 2 \times 5 \times \cos 60^\circ + 5^2 = 5 + 25 = 30$$